On-demand Wakeup Application Note

Revision History

Date	Version	Description
September 29, 2021	0.50	Initial version created.

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1 Overview

This application note describes the system, software architecture, configurations and state transitions of the Wakeup Receiver (WURX) examples running on Atmosic M2/M3 platforms. The WURX feature is developed for the application which requires extreme battery life and has infrequent on-demand activities. In an on-demand wakeup application, the WURX device stays in the hibernation mode with ultra low power consumption most of the time. The device wakes up only when WURX receives a matched wakeup pattern at the specific locations with the wakeup source device deployment.

The Software Development Kit (SDK) supports three fixed wakeup patterns:

- Bluetooth LE high duty cycle connectable directed advertising
- Bluetooth inquiry
- iBeacons

The wakeup pattern can be generated from the BLE_adv_scan example, Atmosic Android app and Atmosic iOS app.

2 WURX System

An On-demand Wakeup system consists of the receiver (WURX device) and the transmitter (wakeup source device). Figure 1 shows the overall system.

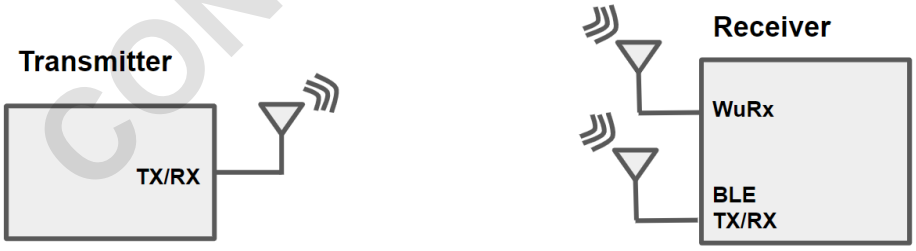


Figure 1 - On-demand Wakeup System Block Diagram

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2.1 Receiver: WURX Device

Ensure two 2.4 GHz antennas are attached to both RFIO and RFIN ports on the ATM2/3 device which is running the WURX application. Figure 2 shows the WURX Evaluation Kit (EVK) setup.

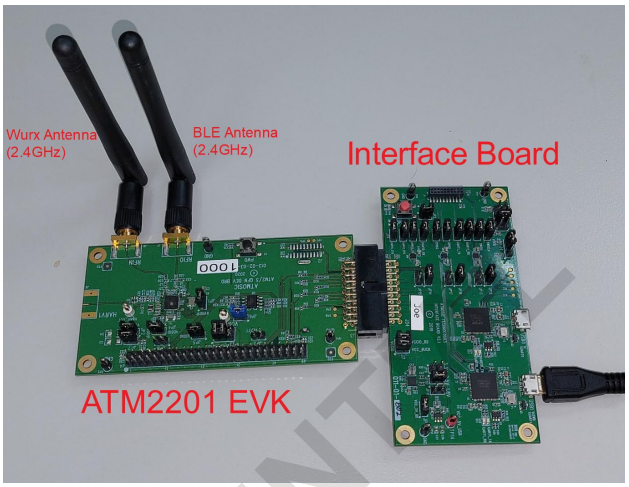


Figure 2 - WURX EVK Setup

2.2 Transmitter: Wakeup Source Device

2.2.1 Bluetooth LE High Duty Cycle Advertiser

To generate high duty cycle advertisements, another ATM2/3 Evaluation Board (EVB) is required. Download the BLE_adv_scan example firmware into the EVB via default configuration build command:

```
[SDK_folder]/atmosic_sdk/platform/atmx[1]/ATM22xx-xx[2]x/example/BLE_adv_scan/
$make run_all
```

2.2.2 Bluetooth Inquirer

There is a WURX function in the Atmosic Android app to activate Bluetooth inquiry. Install the Atmosic app via the following link:
https://play.google.com/store/apps/details?id=com.atmosic.application

The Bluetooth inquiry signals can be generated by other Bluetooth classic only or dual mode devices when those devices are searching for nearby Bluetooth devices.

[1] atm2 or atm3
[2] ATM22xx-x0x, ATM22xx-x1x, ATM32xx-x0x or ATM32xx-x1x

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2.2.3 iBeacon Broadcaster

There is a WURX function in the Atmosic iOS app to start the iBeacon broadcasting which matches the wakeup pattern. Find and install the Atmosic app in the app store.

3 Software Architecture

The SDK includes two WURX examples: WURX_adv and WURX_scan_adv for different WURX application references. Figure 3 shows the hierarchy of these two WURX examples.

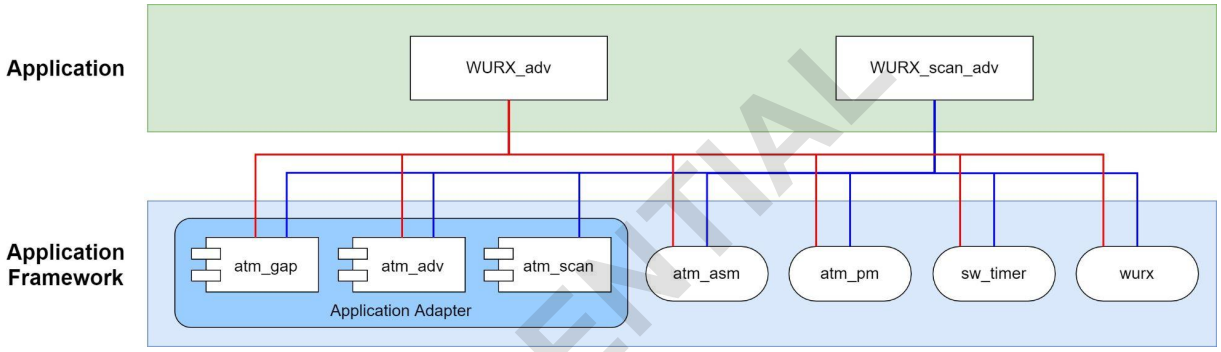


Figure 3 - WURX Examples Hierarchy

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4 WURX_adv Example

4.1 Configurations

WURX_adv supports two build options for different wakeup patterns as shown in Table 1.

Wakeup Pattern	Build Command	Description
BLE High duty cycle advertisement	`$make run_all`	WURX works with the parameter for the high duty cycle advertisement which is stored in the *b4-PMU_WURX* NVDS tag.
Bluetooth inquiry or iBeacon	`$make run_all REF_BCN=WURX_PHONE_ADV`	Firmware alternately uses Bluetooth inquiry and iBeacon configurations for WURX parameters on the fly.

Table 1 - WURX_adv Build Options

Table 2 shows the specific configurable Non Volatile Data Storage (NVDS) tags for WURX applications only.

NVDS tag	TDS File	Description
b4-PMU_WURX	high_duty_adv	PMU WURX parameters
21-APP_WURX_BLE_ACT_CRT_CMD	legacy_nc_ns_30ms	WURX adv. create command parameters
22-APP_WURX_BLE_ADV_DATA	wurx_adv	WURX adv. data
20-APP_WURX_BLE_ACT_STRT_CMD	10_adv	WURX adv. start command parameters

Table 2 - WURX Configurable NVDS Tags

4.2 State Transitions

4.2.1 Default WURX Configuration

Figure 4 shows the state transitions of the WURX_adv example with default WURX configuration (high duty cycle advertisement). ATM2/3 advertises once then enters into hibernation with WURX enabled. Whenever a WURX event is declared, the ATM2/3 wakes up from hibernation and transmits a series of beacons and then returns to hibernation for the next WURX event.

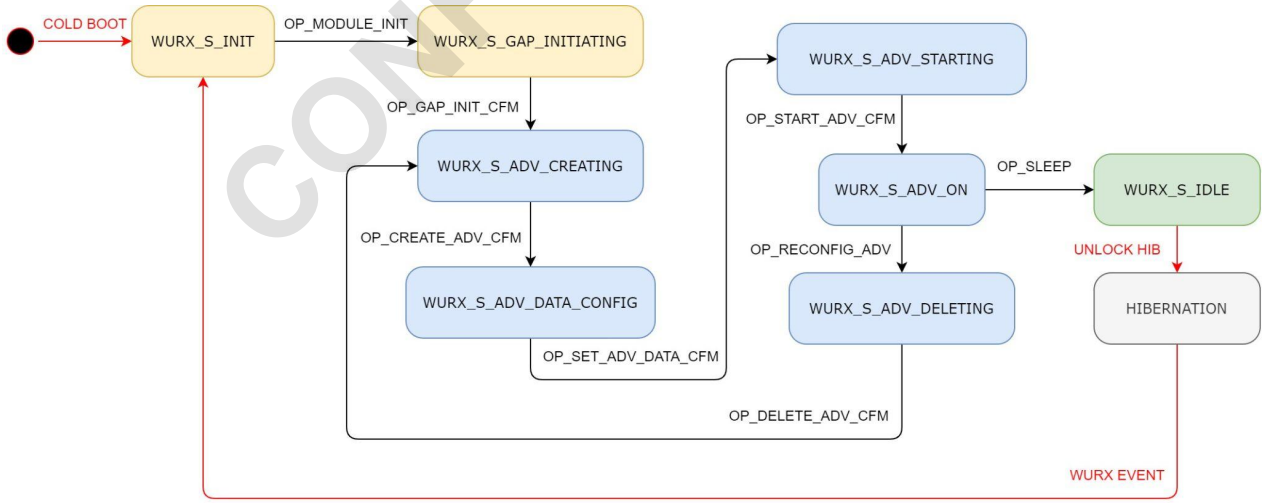


Figure 4 - WURX_adv State Transitions

4.2.2 WURX_PHONE_ADV Configuration

Figure 5 shows the state transitions of the WURX_adv example with WURX_PHONE_ADV configuration which supports both Bluetooth inquiry and iBeacon wakeup patterns alternatively.

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ATM2/3 enters into retention voltage mode with WURX enabled instead of hibernation and wakes up periodically to switch WURX parameters between Bluetooth inquiry and iBeacon configurations and then go back to retention mode. Whenever a WURX event is declared, the ATM2/3 wakes up from retention mode and transmits a series of beacons and then returns to retention mode.

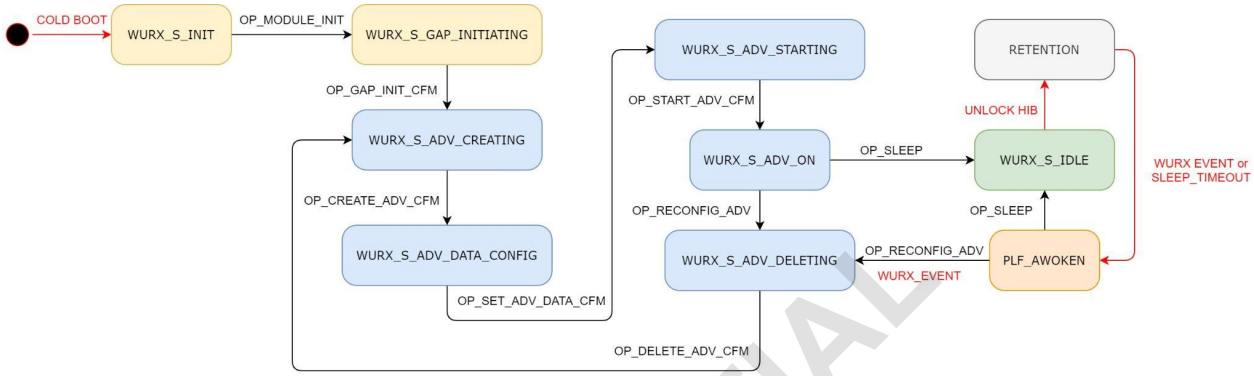


Figure 5 - WURX_adv with WURX_PHONE_ADV Configuration State Transitions

5 WURX_scan_adv Example

5.1 Configurations

The WURX_scan_adv example has only one default build option, but its Bluetooth Device (BD) address must be assigned to a specific address to match with BLE_adv_scan's high duty cycle direct advertisement. Here is the build command to assign correct BD address:

```
$make run_all USER_BD_ADDR:=”C9 11 11 6B 69 7C”
```

Besides, there are some configurable parameters for false wake handling in *a5-WURX_PROFILE* NVDS tag as shown in Table 3.

Parameter	Default Value	Unit	Description
Number of false wake	5	#	A threshold of false wake times to trigger false wake handling: disable WURX and sleep for a certain time to avoid continuous false wake events. The value shall be less than 1024.
Sleep duration	100 (0x64)	cs	The sleep time without WURX enabled when false wake handling is activated.
Long duration	1152 (0x480)	ms	The time to reset the false wake counter.

Table 3 - WURX_PROFILE Parameters

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5.2 State Transitions

This example only works with Transmitters that can send Bluetooth LE high duty cycle directed advertisements. The ATM2/3 advertises once then enters into hibernation with WURX enabled. Whenever a wakeup event is declared, the ATM2/3 scans for directed advertisements that match its BD address. The address “C9 11 11 6B 69 7C” shall be used in WURX_scan_adv 01-BD_ADDRESS NVDS tag, otherwise the adv_param.peer_addr addr parameter in dir_hdc.tds for 06-APP_BLE_ACT_CRT_CMD NVDS tag in BLE_adv_scan example should be modified to match the BD address of WURX device. If found, the ATM2/3 transmits a series of beacons and then returns to hibernation. All of the beacon parameters are programmable via NVDS tags. If no match is found, the ATM2/3 returns to hibernation without beaconing. As a result, the Transmitter wakes up all devices that are in range but only one sends beacons.

This example also tracks the number of consecutive false wakeups. If the number reaches a programmable threshold, the wakeup receiver is disabled and the ATM2/3 is configured to wake up after a programmable time interval. The ATM2/3 stays in this mode for a programmable period of time or until there is a valid targeted wakeup.

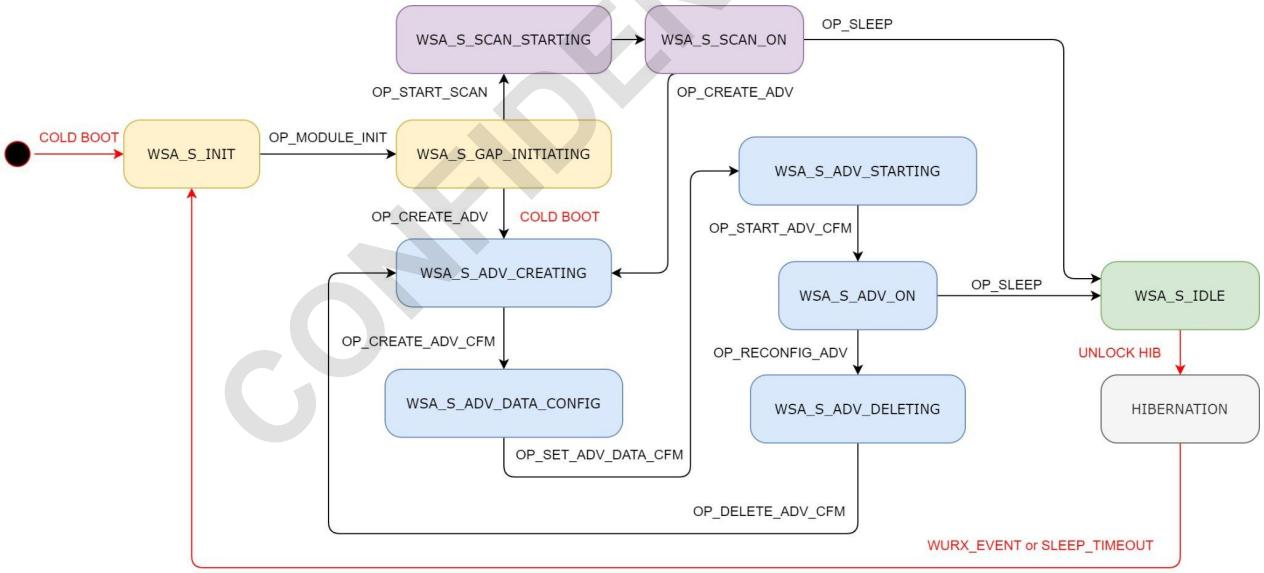


Figure 6 - WURX_scan_adv State Transitions

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